Temperature and particle motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a particle model to distinguish average motion from total energy.

SUCCESS CRITERIA - I can...

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “kelvin” and “particle” to compare your model or result. Sentence starters are optional.

kelvin: the absolute temperature scale used in thermal ratios

particle: a small constituent, such as an atom or molecule

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

An object at 350 K sits in a vacuum facing surroundings at 230 K. How can it cool, and are absolute-zero surroundings required?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

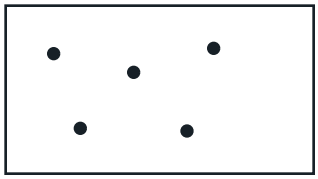
Reference: $K_{avg} \propto T$; $T(K) = T(°C) + 273.15$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

Worked reasoning: In a gas model, warmer temperature corresponds to greater average random molecular kinetic energy. Individual particles still have a range of speeds.

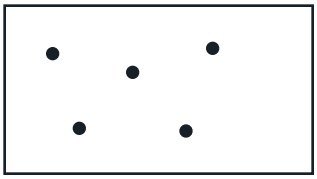
Your turn: Draw two equal-size samples using dots and arrows; label arrows as particle velocities and make the warmer sample's average arrow length greater.

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

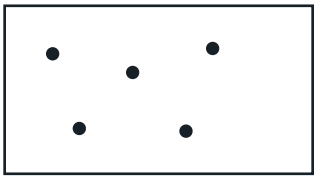
03 Read and interpret the model

Each box represents a sample of the same ideal gas. One is cooler and one warmer. Add particle-velocity arrows with a range of lengths in each box. Make the warmer sample's average kinetic energy greater, without making every particle identical.

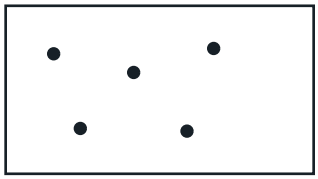
Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

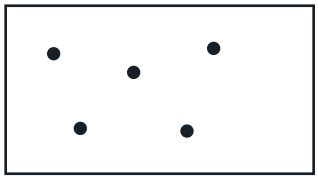
04 Apply the physics

An ideal-gas simulation warms a sample from 300 K to 600 K.
Use average translational energy = (3/2)kT with k = 1.38 × 10^{-23} J/K.
Calculate the average energy per particle at each temperature.
Label the model with the comparison.

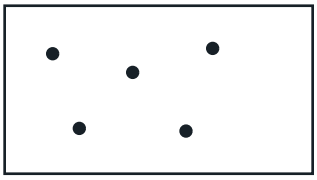
Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student doubles 20°C to 40°C and claims average particle kinetic energy doubles.
Explain why the scale matters.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A constant 40 N forward force acts at steady 6 m/s for 8 s.
Find mechanical power and work.
Use the table to distinguish energy from its transfer rate.

Word help: power = rate of energy transfer

Sentence starter: I found … [unit]. This value means … in this situation.

Case	Energy (J)	Time (s)	Power (W)
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

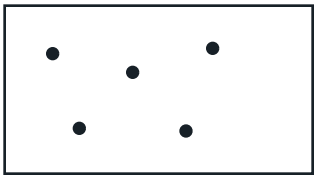
07 Exit check

For a fixed ideal-gas particle mass, average kinetic energy doubles. By what factor does rms speed change?

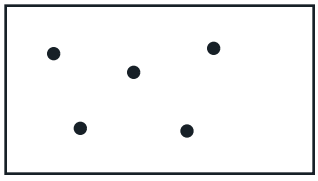
Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

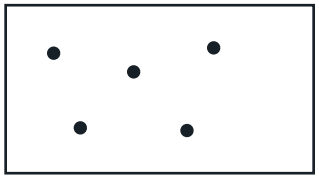
08 STRETCH: test the model

For the same ideal gas, typical molecular speed scales with√T in kelvin 0.
Compare typical speeds at 300 K and 1200 K. Why is a Celsius ratio inappropriate?

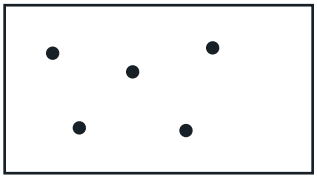
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Sample B



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